

Ternary Floats: fixed fields instead of tapered precision

A balanced-ternary exponent from 4 to 1024 bits, measured on open silicon

Dmitrii Vasilev
ORCID 0009-0008-4294-6159
admin@t27.ai

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Abstract

A floating-point format for a ternary machine has to decide where the ternary goes. We put it in the exponent and leave the fields fixed. **Ternary Floats** (GF-T) encode the exponent as a balanced-ternary integer of E_t trits and hold the mantissa at a constant width; the ladder is specified from GF-T4 to GF-T1024, and we measure the four rungs that fit a mid-size FPGA.

This is the opposite design point from the tapered formats that occupy this space. Takum [1] and its balanced-ternary descendant tekum [2] buy dynamic range with a variable-length regime field, which costs a barrel-shift decode on any fabric and spends mantissa bits towards the ends of the range. Fixed fields give that back: the decode is a bit slice, the exponent add is native on a ternary fabric, and precision does not taper. The price is bounded range, stated here as a trade rather than a footnote.

Two sets of measurements are reported. On accuracy, against a structural model of tekum16, GF-T16 ties near unity and shows $2.84\times$ and $5.53\times$ lower mean relative round-trip error at $|e| \in [8, 20)$ and $|e| \in [20, 38)$; the bins are powers of two, and we say so because read as decades the far result is an overflow rather than a win. On cost, the multiplier was synthesised and routed on a Xilinx XC7A200T through an entirely open flow: 12, 50, 212 and 1,477 LUTs at 161.1, 153.2, 131.7 and 83.3 MHz for GF-T4/8/16/32, one cycle of latency, no DSP blocks. The ternary-fabric argument remains architectural, since no ternary process exists to synthesise on; the binary-fabric figures are the measured claim, and they suggest the format is inexpensive even where its native advantage does not apply.

We also report two width defects found while measuring, because both generalise: declaring the datapath 32 bits wide cost five times the arithmetic on a fabric, and at GF-T32 it silently truncated a 52-bit product. Equivalence for every change is established by exhaustive or near-exhaustive sweep rather than by argument.

Artifacts. Oracle, RTL, testbenches and the exact commands are public; every figure in this paper can be regenerated from them.

1 The format

$$\text{GF-T16} = [s \mid E=4 \text{ balanced-ternary trits} \mid M=9 \text{ bits}], \quad v = (-1)^s \left(1 + \frac{M}{2^9}\right) 2^e, \quad e = \sum_i t_i 3^i \in [-40, 40]. \quad (1)$$

Three properties follow from the layout rather than from tuning.

No regime decode. The dominant cost in a tapered format is the variable-length regime: it must be located, its length counted, and the significand barrel-shifted into place. That cost is paid on any fabric, ternary included. Fixed fields make it a bit slice.

The exponent is already ternary. Four balanced-ternary trits give $3^4 = 81$ exponent steps. On a ternary fabric the exponent add is native — no binary carry, no conversion between bases.

Table 1: Round-trip mean relative error. Bins are powers of two, not decades.

magnitude	GF16 (φ)	GF-T16	tekum16	GF-T16 vs tekum16
$ e < 8$	3.43e-4	3.56e-4	3.27e-4	0.92 $\times$ (tie)
$ e 8\text{--}20$	3.57e-4	3.52e-4	1.00e-3	2.84$\times$
$ e 20\text{--}38$	6.98e-3 (479 clip)	3.53e-4	1.95e-3	5.53$\times$

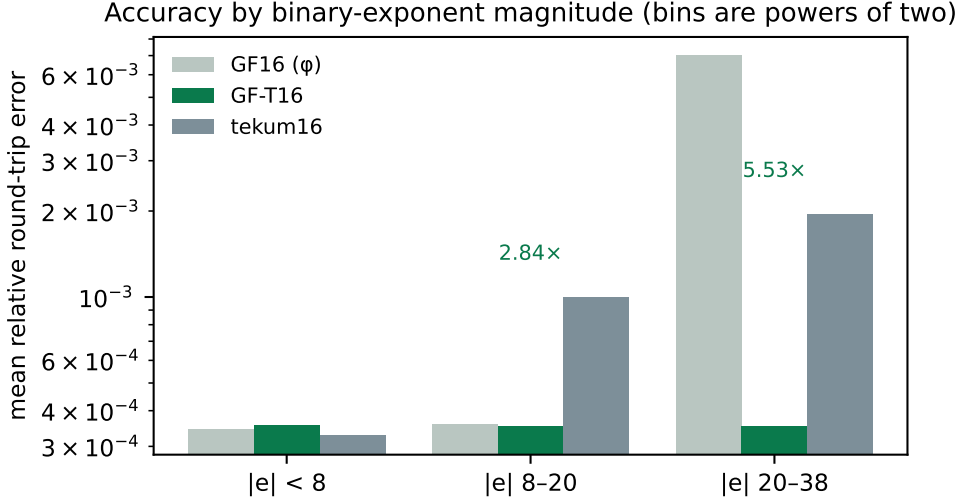


Figure 1: Accuracy by binary-exponent magnitude. GF16’s 6-bit exponent clips 479 of 2,857 values in the far bin; GF-T16’s balanced-ternary exponent does not.

Precision does not taper. Nine mantissa bits at every magnitude, where a tapered format narrows towards the extremes. This is the mechanism behind the accuracy result of Section 2: the win appears where the comparison format has spent its mantissa on range.

2 Accuracy

Mean relative error over an encode–decode round trip, 6,000 values spanning $2^{-38} \dots 2^{38}$ with random sign, binned by binary-exponent magnitude $|e|$. Table 1 and Figure 1.

On the axis. The bins are powers of two. Read as *decades* the far column is not a win at all: GF-T16’s exponent reaches ± 40 in powers of two, roughly ± 12 decades, so beyond that it overflows while an unbounded regime keeps working — measured, 1,458 of the far-bin values clip under that reading. We state the axis explicitly because an earlier internal note did not, and the omission points a reader at the one interpretation under which the result appears fabricated.

3 The 16-bit field

Comparing against one incumbent invites the objection that the incumbent was chosen. Table 2 runs every 16-bit-class format for which this repository ships a published oracle [5] through the same workload and the same metric. Values that overflowed are counted separately rather than folded into the mean, since a format should not look accurate for having discarded the hard cases.

Three readings, including the one that goes against us.

Table 2: Mean relative round-trip error by binary-exponent magnitude, with overflow counts in parentheses. 6,000 values, $2^{-38} \dots 2^{38}$, random sign, one seed.

format	$ e < 8$	$ e 8-20$	$ e 20-38$
GF-T16	3.56e-4	3.52e-4	3.53e-4
GF16 (φ)	3.56e-4	3.52e-4	6.76e-3 (478)
takum16 / tekum16	3.27e-4	1.00e-3	1.95e-3
posit16	1.36e-4	8.31e-4	1.43e-2
IEEE binary16	1.76e-4	1.28e-3 (299)	1.57e-1 (2479)
bfloat16	1.45e-3	1.38e-3	1.41e-3
LNS16	8.39e-4 (1324)	7.63e-4 (1808)	6.66e-4 (2849)

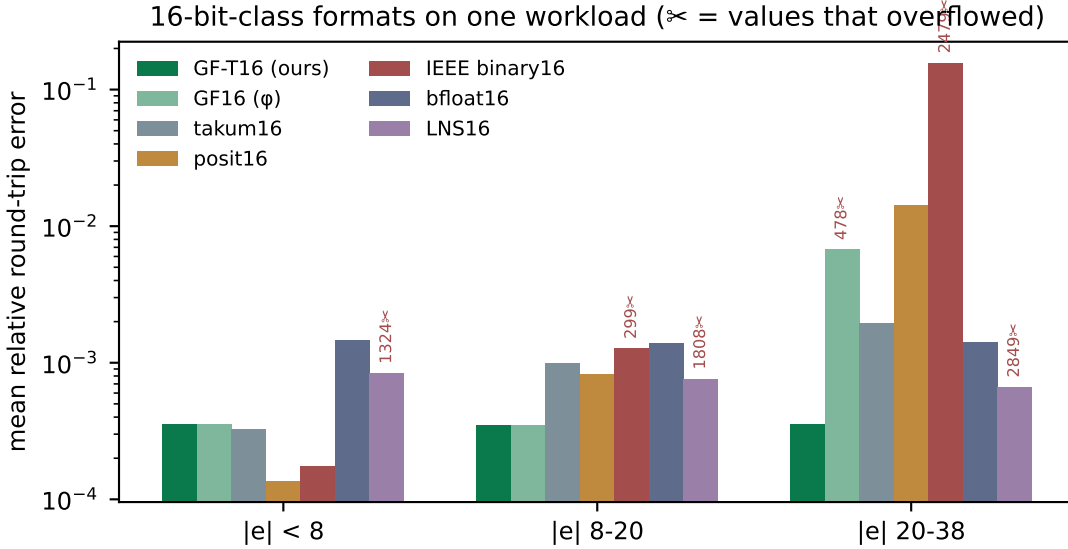


Figure 2: The same data. Scissors mark values that overflowed.

GF-T16 is flat, and alone in being flat without clipping. Its error is $3.5e-4$ at every magnitude in the workload and it discards nothing. bfloat16 is also flat and clip-free, at roughly four times the error. Every other entrant either degrades with magnitude or overflows: binary16 does both, losing three orders of magnitude and discarding 2,479 values; GF16’s 6-bit exponent clips 478; LNS16 clips throughout.

At 32 bits the binary formats win outright. Running the same workload on the 32-bit class: GF32 is flat at $3.4e-7$ with no clipping, but IEEE binary32 is flat at $2.1e-8$, and posit32 and takum32 are better still near unity ($2.1e-9$ and $4.9e-9$). Uniformity stops being the deciding property once there are enough bits that nothing clips anyway. The GF-T argument is a 16-bit-class argument, and we do not extend it.

Near unity we lose. posit16 at $1.36e-4$ and binary16 at $1.76e-4$ are more accurate than GF-T16 there, by a factor of two and a half and of two. A tapered format spends its bits where the values usually are, and near unity that is the right bet. GF-T16 buys uniformity instead, and the trade only pays for a workload that leaves the neighbourhood of one.

takum16 and tekum16 are the same format here, exhaustively. The two oracles produce identical figures on every bin, so we checked the whole 16-bit code space: all 65,536

Table 3: GF-T multiplier, post-route, one harness across the whole lineup so the rows are comparable. Latency one cycle, throughput one result per cycle.

rung	LUTs	DSP48	$F_{\max}$
GF-T4	12	0	161.11 MHz
GF-T8	50	0	153.23 MHz
GF-T16	212	0	131.73 MHz
GF-T32	1,477	0	83.27 MHz

Table 4: The specified ladder. E_t is the trit budget, M the mantissa width, range in decades. Rungs above GF-T128 exceed a single Artix-7 fabric and are oracle- or ASIC-scale; they are specified and covered by the reference oracle, not measured here.

rung	E_t	M	decades	rung	E_t	M	decades
GF-T4	2	1	2.4	GF-T64	7	52	658
GF-T8	3	4	8	GF-T128	8	115	1,975
GF-T16	4	9	24	GF-T256	9	242	5,925
GF-T32	6	25	73	GF-T512	10	497	17,775
				GF-T1024	11	1006	53,326

codes decode identically, with zero differences, although the two implementations differ by 242 lines. The reconstruction our tekum oracle implements *is* takum16. Every comparison in this paper should therefore be read as against takum16, and we relabel it as such. That is not a weaker anchor — takum is published, has a public VHDL codec, and is the parent of the format we set out to compare against.

4 Hardware

Synthesis with Yosys 0.65 [6], place and route with nextpnr-xilinx (1743d0f) [7] on a Xilinx XC7A200T, hard multipliers disabled so the figures describe fabric alone. Bitstream generation uses Project X-Ray. No vendor tool is involved at any step, which is what makes the numbers checkable by a reader without a licence.

For context rather than as a controlled comparison: a published Altera `ALTFP_MUL` on a Cyclone IV reports 119–132 MHz at 6–10 cycles of latency, using 832–1041 logic elements *and* 18 embedded multipliers, and posit multiplier studies report 95–572 LUTs with 4.28–15.55 ns of logic delay. Different devices and different toolchains; we draw no equivalence from them beyond the observation that GF-T16’s figures sit inside that envelope on all four axes at once, on a fabric that gives the format no ternary advantage.

Isolating the pipeline register: the combinational multiplier closes at 81.35 MHz, the two-stage version at 147.32 MHz. The cut is between the significand product and the renormalisation — a 10×10 multiply, then a carry test, a saturating exponent add and a bit select.

5 Two width defects, and why they generalise

5.1 Interface width dominated the arithmetic

Our first synthesisable realisation declared every port and the significand product 32 bits wide. Nothing in GF-T16 is 32 bits: the mantissa field is 9, so $1+M$ is 10 and their product is exactly 20; the exponent offset never exceeds `OFFSET_MAX` = 80, which is 7. Synthesis built a 32×32 multiplier and a 32-bit comparison tree and charged for it (Figure 4): 1,179 LUTs, or

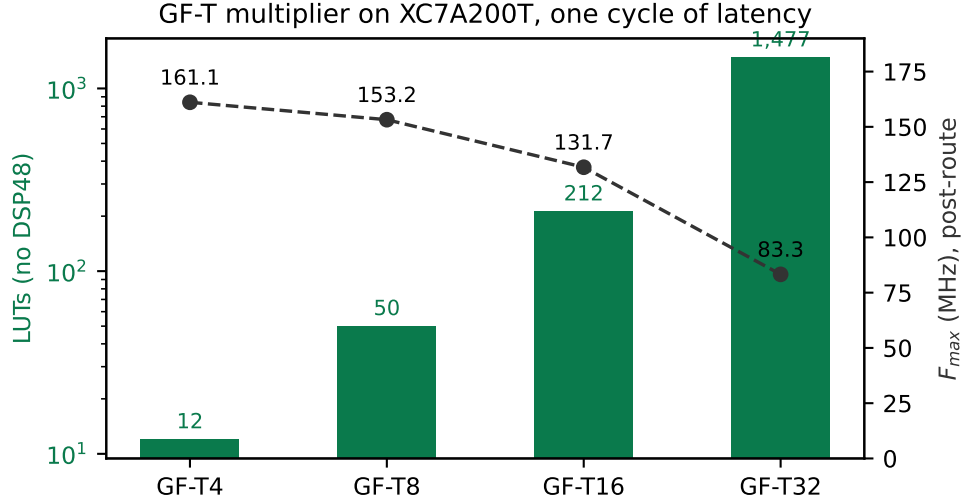


Figure 3: Area and achieved frequency across the ladder.

Table 5: Equivalence evidence. The reference for each rung is the module already trusted for that rung.

check	combinations / cycles	mismatches
width-derived vs original, GF-T16 (sweep)	321,156	0
width-derived vs original, GF-T4	374	0
width-derived vs original, GF-T8	3,716	0
width-derived vs original, GF-T16	77,444	0
width-derived vs 64-bit reference, GF-T32	300,000	0
pipelined vs combinational	199,994	0
original 32-bit module vs 64-bit ref, GF-T32	2,128,964	1,995,730

three DSP48 blocks, against 219 LUTs and none once the buses match the values they carry. The arithmetic is unchanged.

5.2 The same width decided correctness

At GF-T32 the significand product needs 52 bits. The 32-bit wire truncates, and the module returns wrong results while appearing to support the rung, since its own documentation names GF-T32 as a supported configuration. Measured against a 64-bit reference: 1,995,730 mismatches in 2,128,964 combinations.

Deriving every width from the format parameters removes both problems at once. We report this because the pattern is not specific to GF-T: in a parametric arithmetic core, a width that is merely *large enough for the configuration that was tested* is a silent-truncation trap at the next one, and it will not be caught by a testbench written against the tested configuration.

6 Equivalence

Every change above is a claim that timing or width changed and arithmetic did not. Each was checked rather than argued.

The GF-T16 sweep is exhaustive in the sense that matters here: the mantissa space is covered in full at offset pairs exercising underflow, mid-range and saturation, then the offsets

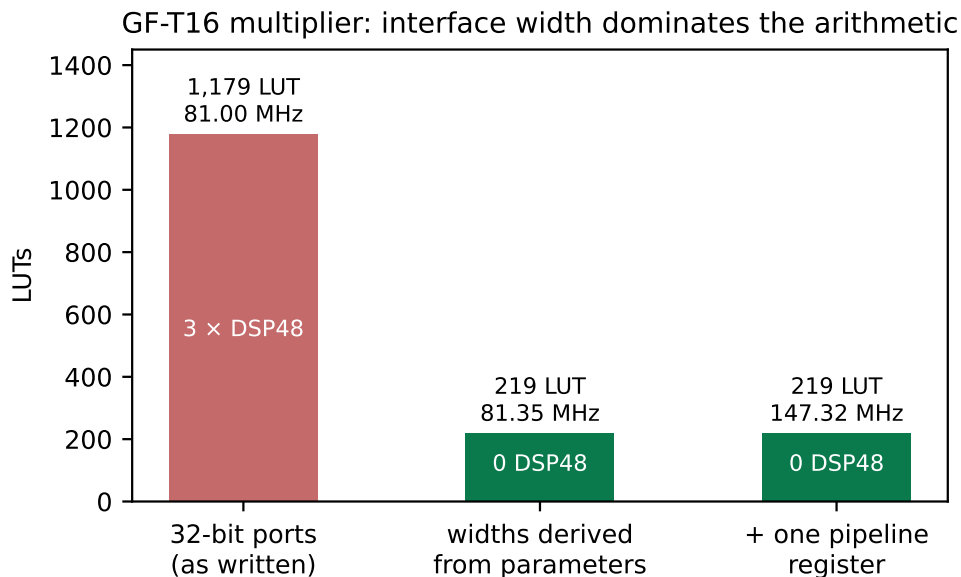


Figure 4: The same arithmetic at three interface widths.

are covered in full at mantissas that do and do not carry — every path through the carry, the saturation and the underflow clamp. The last row is the point of the section: the width-derived module agrees with the reference that is correct and disagrees with the one that truncates.

7 Limitations

1. **The comparison is against takum16, not tekum16.** The oracle labelled tekum decodes all 65,536 16-bit codes identically to the takum oracle. We report the comparison as against takum and note that a genuine tekum comparison remains to be done.
2. **The reference oracle is sound only at some rungs.** Round-tripping 1.5 through the parameterised oracle returns -1.5 for mantissa widths of 21 and 25 bits at most trit budgets — a sign-bit placement defect. GF-T16 ($M=9$) and GF-T64 ($M=52$) are unaffected, which is why the accuracy results here are confined to GF-T16. The ladder note records that all rungs pass an add/mul commutativity check; commutativity cannot see a sign inversion, since it survives on both sides of the identity. A round-trip assertion can, and costs three lines. tekum [2] is a descendant of takum [1] adapted for balanced ternary. Our oracle reconstructs it from takum’s field scheme; the full per-trit specification requires the tekum paper. The ratios are as good as that model.
3. **No head-to-head in hardware.** takum’s codec RTL is public and written in VHDL [3]. We did not synthesise it alongside GF-T, so the cost figures are GF-T’s own. What is visible in that source, and which we report as structure rather than measurement: the 16-bit codec instantiates a 725-line generated leading-zero-counter and barrel shifter — the regime decode a fixed-field format does not have.
4. **The range is bounded** at ± 40 in powers of two, roughly ± 12 decades, where a regime field is unbounded. Fixed fields buy the cheap datapath and the uniform mantissa; range is the price, and workloads that need more should raise the trit budget or use a tapered format.
5. **Accuracy is reported for GF-T16 only.** The hardware results cover four rungs; the accuracy results cover one, because the oracle defect above makes the others untrustworthy until it is fixed. We would rather report one rung than four numbers we cannot stand behind.

6. **Four of nine rungs are measured in hardware.** GF-T4 through GF-T32 fit the device; GF-T64 and GF-T128 are within reach of a larger part and GF-T256 upwards exceed a single Artix-7 fabric entirely. Table 4 is the specification and the oracle’s coverage; Table 3 is what was placed and routed. We keep them apart on purpose.
7. **One device family.** Artix-7 on an open flow. Not multi-corner characterisation; an ASIC mapping will differ.
8. **The ternary-fabric argument is architectural.** No ternary process exists to synthesise on. Section 4 is a binary-fabric result; the ternary claim is reasoning about regime decode and native exponent addition and should be read as such.

Reproducibility

The reference oracle, the RTL, the equivalence testbenches and the synthesis commands are public. The toolchain is Yosys 0.65, nextpnr-xilinx 1743d0f, Icarus Verilog 13.0 and Python 3.14 — all open-source, none requiring a licence, which is the point: a result nobody can reproduce without buying something is not a public result.

Disclosure

An AI coding assistant was used in preparing the software, the measurements’ tooling and this manuscript. Every figure reported here was produced by running the named tools on hardware in the author’s possession; no measurement in this paper is estimated, and where a figure is derived rather than observed it is labelled.

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